

WELL-EQUIPPED

John Dee was an expert at navigating by the sun and stars, with the help of specialized equipment. This astronomical compendium incorporates an ingenious range of instruments to help guide the course of a ship, including a compass, a wind vane, and a sundial.



1625 engraving of an alchemist

THE SCIENCE OF THE STONE

Dee was also interested in alchemy, the scientific search for the magical "philosopher's stone," which could turn metals such as lead into gold. Ben Jonson made fun of this practice in his 1611 comedy *The Alchemist*. Jonson's alchemist, Subtle, is a fraud who cheats greedy and gullible people out of their money using lies and trickery.



Sundial and compass

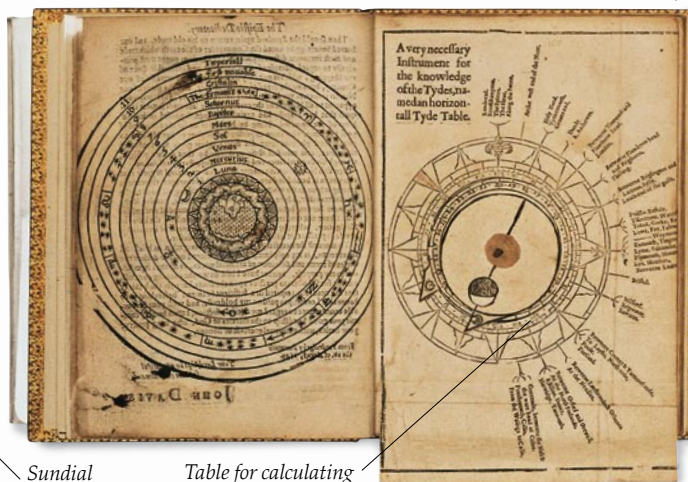
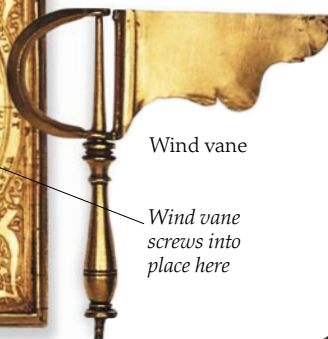


Table for calculating tides from the phases of the moon



Wind vane

Wind vane screws into place here

BEST BOOK

In 1595, scientist John Davis (c. 1550–1605) wrote *The Seaman's Secrets*, the most accurate guide to navigation of the 16th century. Davis was a skilled navigator and an experienced explorer. He made three voyages into the Arctic, hoping to find a route to China. Davis was eventually killed by pirates while on route to Sumatra, Indonesia.



A replica of Galileo's telescope

Galileo's telescope was a great improvement on a Dutch invention

BACK TO THE SUN

John Davis invented the back-staff, an instrument for determining a ship's latitude, or north-south position, from the height of the sun. Earlier instruments, such as the cross-staff, forced navigators to stare into the sun. With a back-staff, navigators could turn their backs to the sun and use a shadow to measure its height.

Davis's back-staff, was the most popular navigation instrument until well into the 17th century

SECRET STUDIES

John Gielgud starred as scientist-magician Prospero in the 1991 movie *Prospero's Books*, which was based on Shakespeare's *The Tempest*. Prospero is a mysterious character, who describes himself in the play as being "rapt in secret studies." Shakespeare created him at about the same time that Jonson was writing *The Alchemist*.



The navigator looked through the slit in the sighting vane

The sighting vane was positioned at the estimated latitude

"These late eclipses of the sun and moon portend no good to us."

WILLIAM SHAKESPEARE
The Earl of Gloucester in *King Lear*